

Life Expectancy Changes Due to Kidney Donation Across Race, Sex, and Age

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Abstract

Background. Living kidney donors face a small but well-documented elevation in long-term risk of end-stage renal disease (ESRD), yet this risk has not been translated into a decision-relevant life-expectancy metric stratified by the demographic factors known to modify it.

Methods. I constructed a Markov microsimulation model comparing lifetime outcomes for matched donor and non-donor cohorts, entering at the age of donation and simulated in annual cycles, parameterized from the United States Renal Data System (USRDS), the Scientific Registry of Transplant Recipients (SRTR), and published cohort studies of post-donation ESRD incidence and mortality.

Results. In the base case, donation was associated with a mean life-expectancy cost of 911 days (2.5 years). This cost was slightly larger for Black donors than White donors and roughly 90% larger at age 25 than at age 55. The result depends heavily on published estimates of the donor all-cause mortality hazard ratio (HR), which disagree not because they measure the same quantity with noise, but because the largest meta-analysis and the longest-follow-up cohort study describe different points on what is plausibly a non-proportional (time-varying) hazard, with excess mortality emerging only after approximately a decade. Holding the HR fixed at any single published value shifts the base-case cost from a net benefit of several years ($HR = 0.60$) to a cost of more than four years (the upper confidence bound of the long-follow-up estimate). Any other modeled parameter moves the result by a maximum of about six weeks. The life-expectancy value of prior-living-donor priority waitlist access and of a National-Kidney-Registry-style donation voucher was negligible unconditionally (under one day) but worth over one hundred days conditional on the designated beneficiary's actually developing ESRD.

Conclusions. These findings support age- and race-specific life-expectancy disclosure during living-donor informed consent, indicate that priority-access mechanisms function as catastrophic insurance rather than as interventions expected to shift average outcomes, and highlight resolving the time-course of donor all-cause mortality risk as a research priority, since it dominates every other source of uncertainty in this analysis.

Keywords

living kidney donation, life expectancy, Markov model, end-stage renal disease, health disparities

Highlights

- Kidney donation costs the average donor about two and a half years of life expectancy.
- This cost is slightly larger for Black donors than White donors, and roughly 90% larger at age 25 than at age 55.
- This estimate depends heavily on an unresolved question in the literature: whether donor all-cause mortality risk rises only after roughly a decade post-donation, or not at all. Depending on which published estimate is used, the result ranges from a net benefit to a loss of several years.
- Priority transplant access for donors and their designated family members is worth under one day on average, but 100–500 days if kidney failure actually occurs.

Introduction

Living donor kidney transplantation (LDKT) remains the treatment of choice for end-stage renal disease (ESRD), offering better graft function and longer allograft survival

than deceased-donor kidney transplantation (DDKT).¹ Yet the gap between organ need and organ supply persists: more than 4,000 candidates died on the US kidney waiting list in 2024, and a comparable number were removed for transplant-disqualifying deterioration, while the annual number of LDKTs has remained essentially flat for over a decade.² Expanding living donation therefore rests on two complementary levers: (1) recruiting more donors and (2) better understanding, quantifying, and managing the risk that donation poses to the donor.

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The safety literature on living donation is large, but has historically emphasized relative rather than absolute, or life-expectancy-scaled, risk. Perioperative mortality is low (3.1 per 10,000 donors) but not uniform, rising sharply with male sex, Black race, and pre-existing hypertension.³ Longer-term follow-up using healthy, contraindication-screened non-donor comparators established that living donors face a several-fold higher risk of subsequently developing ESRD themselves, even though the absolute magnitude of that risk remains small.^{4,5} Whether this translates into excess all-cause mortality is unresolved, and the disagreement in the literature is not simply a matter of noisy estimates of one fixed effect.

To date, O’Keeffe et al. (2018)⁶ completed the largest meta-analysis of follow-up studies of long-term mortality among living kidney donors. They pooled four cohort estimates among donors relative to healthy matched controls: Segev et al. (2010)³, with a 6.3-year follow-up; Garg et al. (2012)⁷, with a 6.5-year follow-up; Berger et al. (2011)⁸, with a 10-year follow-up; and Mj  en et al. (2014)⁹ with a 15.1-year follow-up. Of those four studies, the only one to report a hazard ratio (HR) greater than 1.0 was that of Mj  en et al. (HR= 1.30, 95% CI 1.11–1.52). O’Keeffe et al. concluded “We found no evidence that living kidney donors had higher risk for mortality.” Mj  en disagreed with this conclusion, arguing in a letter responding to O’Keeffe’s meta-analysis: “the survival curves in our study only started to separate at more than 10 years of follow-up, indicating that increased mortality after kidney donation may manifest only after long-term follow-up. At a follow-up of less than 10 years, the findings from these...studies are not contradictory. No published studies on kidney donors oppose our finding of increased mortality only after more than 10 years of follow-up.”¹⁰ O’Keeffe agreed.¹¹ No detectable excess mortality within the first post-donation decade, alongside a significant excess in the one cohort followed long enough to observe it, is consistent with a time-varying hazard rather than a single fixed effect that the literature estimates with varying precision, and is treated accordingly in this analysis.

Post-donation ESRD risk is not distributed evenly across the donor population. Black donors experience roughly three times the 15-year cumulative incidence of ESRD observed in White donors, a disparity that persists after matching to equally healthy non-donor comparators of the same race.⁴ Part of this excess risk has a defined biological correlate: two-copy *APOL1* renal-risk-variant genotypes, disproportionately carried by Black individuals, are among the strongest known risk factors for progression to chronic kidney disease (CKD) stage 3 or higher and have been directly implicated in the long-term kidney health of Black living donors.¹² Donation itself is also not evenly distributed: women have consistently made up a majority of living donors, a pattern attributable in part to financial barriers to donation and disparities in access to unpaid leave that fall more heavily on lower-income men.¹³ Together, these findings motivate stratifying any life-expectancy analysis of donation by age, race, and sex rather than reporting a single population-average effect.

Policy has responded to donor risk primarily by protecting donors’ future access to a transplant should they themselves develop ESRD. The Kidney Allocation System implemented

in 2014 (KAS250) grants prior living donors priority access to the deceased-donor waiting list, and early evaluation confirmed that this priority meaningfully shortened prior donors’ time to listing.¹⁴ Independently, a national registry analysis of transplant candidates who had previously donated an organ found that such priority access translated into substantially shorter waiting times and higher-quality kidneys compared with matched non-donor candidates, confirming that the policy achieves its intended effect at the level of the individual recipient.¹⁵ The National Kidney Registry (NKR) has extended this priority concept beyond the donor’s own future risk through its voucher program, under which a donor may designate family members, or in some cases any future intended recipient, to receive expedited, priority access to a living-donor kidney should they ever need one. Two large Monte Carlo simulation studies, spanning 50- and 100-year horizons, have shown that the voucher program is programmatically sustainable.^{2,16}

These two literatures, quantifying the donor’s own excess ESRD and mortality risk and demonstrating the programmatic success of priority-access mechanisms, have so far remained largely separate. The voucher simulations answer a supply-side question (will there be enough donors to honor future redemptions?) but do not quantify the demand-side question of how much life expectancy a voucher, or PLD priority access more generally, is actually worth to the person who holds it. Similarly, the risk literature reports relative risks and cumulative incidences but stops short of translating these into an integrated, individual-level life-expectancy trade-off that simultaneously accounts for the donor’s elevated risk and demographic heterogeneity. This paper addresses that gap using a Markov microsimulation model that estimates the change in life expectancy attributable to kidney donation itself, stratified by age at donation, race, and sex, and separately quantifies the life-expectancy value of priority waitlist access, both unconditionally at the population level and conditional on the recipient having actually developed ESRD.

Methods

Model overview

I constructed a Markov state-transition model to compare lifetime outcomes for two hypothetical cohorts of otherwise identical individuals: one cohort that donates a kidney and one that does not. Both cohorts enter the model at the age of donation in full health, and the simulation advances in annual cycles until age 110. The analysis was conducted from the perspective of the individual donor rather than the health-care system; outcomes are reported as undiscounted remaining life-years, consistent with the convention in burden-of-disease analyses.¹⁷ The base-case age at donation is 40 years; subgroup analyses span ages 25–55. I used a Monte Carlo simulation with 5×10^6 individuals per arm and cross-validated against a deterministic analytic cohort implementation. I used annual (one-year) cycles throughout. The Markov model structure is shown in Figure 1.

Health states

The model contains seven mutually exclusive health states (Figure 1): *Healthy*, *ESRD onset*, *Dialysis*, *Standard waitlist*, *Priority waitlist*, *Post-transplant*, and *Dead* (absorbing). At ESRD onset, the cohort branches: a fraction is listed preemptively (bypassing dialysis entirely) and the remainder initiates hemodialysis. The donor arm routes ESRD survivors to the *Priority waitlist*, reflecting the allocation priority afforded to prior living donors (PLD) under KAS250; the non-donor arm routes to the *Standard waitlist*. Graft failure returns patients to the *Dialysis* state, bypassing the ESRD-onset branch.

ESRD risk after donation

Fifteen-year cumulative ESRD incidence was taken from Muzaale et al.⁴: 30.8 per 10,000 (95% CI 24.3–38.5) for donors and 3.9 per 10,000 (95% CI 0.8–8.9) for age-, sex-, and health-matched controls (8-fold relative risk). Race-specific donor rates were 74.7/10,000 (Black) and 22.7/10,000 (White).⁴ The White non-donor baseline was taken from Grams et al.¹⁸ (sex-averaged 0.05%/15 yr), because zero ESRD events were recorded in White matched controls in Muzaale et al.

I modeled ESRD hazard with a Weibull distribution. The shape parameter ($k = 2.54$) was fit, rather than assumed, from the five cumulative-incidence-of-ESRD curves (median, interquartile range, and 1st/99th percentiles) that Massie et al. report at 5, 10, 15, and 20 years post-donation.⁵ The Weibull CDF linearizes under the complementary log-log transform, $\ln(-\ln(1 - I(t))) = k \ln(t) - k \ln(\lambda)$, so ordinary least-squares regression of the transformed incidence on $\ln(t)$ recovers k directly as the slope; applied to the median curve this gives $k = 2.54$, reproducing all four published points to within 5%. Fitting the same regression separately to each of the other four published curves gives k ranging 2.15–2.54, confirming a single accelerating Weibull shape ($k > 1$) is broadly consistent across the reported risk distribution; a constant-hazard ($k = 1$) model is ruled out, as it would predict only $\approx 21/10,000$ at 20 years from the observed 15-year rate of 16/10,000, versus the 34/10,000 actually reported. Once fit, k was combined with a per-subgroup Weibull scale λ , calibrated via a competing-risk bisection algorithm to reproduce each subgroup's 15-year cumulative incidence while accounting for background-mortality attrition in the at-risk pool. I drew within-donor subgroup hazard ratios from Massie et al.: 2.96 (95% CI 2.25–3.89) for Black race, 1.88 (95% CI 1.50–2.35) for male sex, and 1.40 per decade (non-Black donors).⁵

Dialysis survival and waitlist listing

First-year hemodialysis mortality was 22% and subsequent-year mortality 17% ($\approx 42\%$ five-year survival), consistent with USRDS estimates.¹⁹

At ESRD onset, a fraction of patients is listed preemptively before dialysis initiation, avoiding dialysis mortality entirely. I drew preemptive listing probabilities from USRDS 2025 Annual Data Report (ADR) Figure 7.13: 5.8% overall for the non-donor arm and 9.4% for the donor arm, using the age 18–44 population rate as a proxy for

this younger, healthier, donor-like cohort (USRDS does not report preemptive-listing rates stratified by donor status).²⁰

Among patients on dialysis, the conditional annual probability of joining the waitlist was set to 15%/year (base case), derived from USRDS 2025 ADR Figure 7.17.²⁰ Approximately 25% of 18–44 year-old ESRD patients are listed or transplanted within one year, of whom $\approx 9.4\%$ were preemptively listed, yielding an estimated post-dialysis listing rate of $\approx 15.6\%$ /year for the donor-like cohort.

Transplant waiting time and waitlist outcomes

Transplant probability while listed was modeled as a time-homogeneous exponential process with rate $r = 1/\mu$, where μ is the mean waiting time, the natural parameterization of the exponential distribution.²¹ For the standard waitlist, I used the post-KAS250 national mean of 1,765 days.²² For the PLD priority waitlist, I used the post-KAS median of 102.6 days,¹⁴ converted to a mean via $\mu = \text{median} / \ln 2 \approx 148$ days. Waitlist mortality was 5.0 deaths per 100 patient-years (SRTR 2023 ADR Figure KI 24). Competing removal (delisting for reasons other than transplant or death) was derived from the 3-year cumulative removal incidence of 19.06% reported in SRTR 2023 ADR Figure KI 22, back-calculated to a cause-specific annual rate of 11.4%/year via competing-risk bisection accounting for simultaneous attrition from transplantation and death.¹

Post-transplant survival

I derived post-transplant annual mortality from age-stratified 5-year Kaplan–Meier estimates for DDKT recipients in the SRTR 2023 ADR (Figure KI 70, 2016–2018 cohort): 0.9%/year (ages 18–34), 1.8%/year (35–49), 3.9%/year (50–64), and 6.9%/year (≥ 65).¹ The base case uses DDKT survival because prior living donors who develop ESRD predominantly receive transplants via the deceased-donor priority pathway rather than living-donor transplantation, consistent with USRDS data showing that living-donor transplants account for a small minority of transplants in this population.²³ A one-way sensitivity analysis substitutes LDKT patient survival (SRTR 2023 ADR Figure KI 76: 0.4%, 0.8%, 1.7%, and 3.9%/year by age band) to bound the effect of this assumption. Annual graft failure post-year 1 was likewise age-stratified, derived as $1 - (S_5/S_1)^{1/4}$, where S_n denotes the n -year graft survival probability, from age-specific 1- and 5-year DDKT graft survival in the SRTR 2023 ADR (Figure KI 53, 2016–2018 cohort): 3.9%/year (ages 18–34), 3.4%/year (35–49), 4.9%/year (50–64), and 7.8%/year (≥ 65).¹ Graft failure returned patients to the Dialysis state.

Background mortality and donor all-cause survival

I took background mortality in the Healthy state from the 2021 US Life Tables,²⁴ with sex-specific tables used in sex-stratified analyses. As discussed in the Introduction, the published literature on donor all-cause mortality does not consist of noisy estimates of one fixed hazard ratio (HR): short-to-medium-follow-up cohorts and the meta-analysis pooling them report no excess mortality,⁶ while the one cohort followed long enough to plausibly detect a

late-emerging effect reports a significantly elevated HR,⁹ with survival curves separating only after approximately ten years.^{10,11} I therefore modeled the donor all-cause mortality HR as time-varying rather than fixed: flat at 1.0 (no excess risk) for the first 10 years after donation, then increasing linearly to 1.30 by 15 years post-donation, matching the Mj  en et al.¹⁰ cohort’s median follow-up, and held flat at 1.30 thereafter. The non-donor arm’s background mortality HR is fixed at 1.0 throughout. Because this reflects a genuine question about model structure (does excess mortality risk exist at all, and if so, when does it emerge) rather than sampling variation around an agreed-upon value, it is not included in the probabilistic sensitivity analysis; instead, I report a dedicated structural sensitivity analysis over the plausible alternative HR assumptions in the literature.

Subgroup analyses

I repeated the analyses at ages 25, 35, 40, 45, and 55 years at donation. Donor ESRD rates were scaled by the Massie 2017 age HR (1.40/decade, non-Black donors);⁵ I held non-donor baselines at the Muzaale overall matched-control rate.

I took race-specific donor ESRD rates (Black 74.7/10,000; White 22.7/10,000) and race-specific non-donor baselines (Black sex-averaged 0.195%/15 yr; White sex-averaged 0.05%/15 yr) from Muzaale et al.⁴ and Grams et al.,¹⁸ respectively. Race-specific waitlist mortality and post-transplant survival were from SRTR 2023 Figures KI 25 and KI 71.¹

I derived sex-specific donor ESRD rates by applying the Massie 2017 male-sex HR (1.88)⁵ to the overall donor rate, weighted by the observed donor sex mix (63.5% female; SRTR 2023 ADR Figure KI 104).¹ For non-donor baselines, I used the Grams 2016 White sex-specific rates (female 0.04%/15 yr; male 0.06%/15 yr)¹⁸ rather than applying the within-donor sex HR to the control arm, because Massie 2017 hazard ratios describe variation within the donor cohort and are not appropriate for scaling a different reference population. I applied this sourcing rule, Muzaale matched-control rates for overall analyses and Grams direct rates wherever sex or race is stratified, consistently across all simulations.

Probabilistic sensitivity analysis (PSA)

I conducted a PSA with 2,000 parameter draws. Sampled quantities and their distributions were:

- ESRD 15-year cumulative risks: beta distributions parameterized from the Muzaale 2014 published 95% CIs;⁴
- Weibull shape k : log-normal, $\sigma = 0.069$, the sample standard deviation of $\ln k$ across the five independent cloglog-regression fits described above (Massie et al. does not report a CI for this parameter directly); and
- Waitlist listing probability: beta, fractional SE = 40%, consistent with the 0.05–0.30 plausible range

Registry-derived parameters (from SRTR or USRDS) have negligible sampling error relative to structural uncertainty and were fixed at base-case values; scenario variation in these parameters is captured by the one-way sensitivity analysis. The donor all-cause mortality HR is deliberately excluded

from the PSA: as described above, the disagreement among published estimates reflects a structural question about the time-course of donor risk, not sampling error around a single value, and is instead addressed by the dedicated structural sensitivity analysis below (“Structural sensitivity: donor mortality hazard ratio”). I report the mean Δ LE, 95% credible interval, and the probability that donation is life-expectancy–neutral or beneficial.

One-way sensitivity analysis

I varied each parameter individually, holding the donor all-cause mortality HR at its base-case time-varying profile:

- No priority access and changing the PLD wait time
- ESRD relative risk from the Mj  en upper-bound on ESRD HR
- Post-transplant survival at LDKT quality
- Dialysis mortality $\pm 50\%$

Structural sensitivity: donor mortality hazard ratio

Because the donor all-cause mortality HR is the single largest determinant of the model’s result (see Results) and is not sampled in the PSA or perturbed in the one-way sensitivity analysis above, I separately compared the base-case time-varying HR profile against the flat (time-invariant) HR implied by each individual published estimate, taken at face value for the donor’s entire remaining lifetime: HR = 0.60 (the O’Keeffe et al. pooled point estimate),^{6,11} HR = 1.0 (no excess mortality), HR = 1.30 (the Mj  en et al. point estimate),⁹ and HR = 1.52 (the upper bound of the Mj  en et al. 95% CI). This analysis is intended to bound how much the result depends on which single-HR assumption is adopted, rather than to imply that any of these flat alternatives is itself a plausible model of donor risk.

Calibration

I validated the model’s transplant probability against the SRTR 2023 3-year waitlist outcome distribution (Figure KI 22, 2018–2020 listing cohort).¹ This comparison is non-circular because I derived the transplant probability from the mean waiting time, not from the 3-year outcome distribution. I back-calculated the waitlist removal rate via a competing-risk bisection algorithm to reproduce the SRTR-observed 19.1% removed at three years (11.4%/year). The model over-predicted three-year waitlist mortality (10.5% vs. 6.8% observed, $\Delta = +54\%$), likely reflecting the application of a 2023 cross-sectional mortality rate (confirmed 5.0/100 patient-years from the raw SRTR data file) to a 2018–2020 listing cohort; this bias is symmetric across both arms and conservative (slightly overstates the cost of the waitlist period for both donors and non-donors equally). The model predicted 38.3% transplanted at three years versus 44.3% observed (DDKT + LDKT combined). No independent cohort exists tracing the complete pathway from living donation through ESRD onset to transplantation; this check represents the available face-validity evidence for the model.

Voucher benefit

A kidney-donation “voucher” is a mechanism by which a living donor designates a healthy, non-donor family member to receive PLD priority-waitlist access should that family member ever develop ESRD. I simulated two arms of healthy non-donors, both carrying non-donor baseline ESRD risk and a background all-cause mortality HR of 1.0; the sole difference between arms was waitlist priority upon ESRD onset (voucher arm: PLD mean wait 148 days, derived from the Wainright et al. post-KAS median of 102.6 days via $\mu = \text{median} / \ln 2$; ¹⁴ control arm: standard mean wait 1,765 days ²²). Consistent with the sourcing rule used for the base-case preemptive-listing parameters, I assigned the voucher arm the informed/monitored preemptive-listing probability (9.4%) and the control arm the general-population probability (5.8%). ²⁰ All other transition probabilities (dialysis mortality, per-cycle waitlist listing, waitlist mortality and removal, post-transplant survival, graft failure) were identical to the base-case model. I ran the analysis at the base-case age of 40 with an age sweep (25–60 years) and an ESRD-risk subgroup analysis using the Muzaale overall and Grams race-specific non-donor cumulative incidences. ^{4,18} One-way sensitivity analyses varied the voucher and standard median wait times, the per-cycle listing probability, the preemptive-listing probability, and post-transplant graft quality (DDKT vs. LDKT).

ESRD-conditional priority benefit

Because the voucher (and, by extension, the donor PLD-priority) benefit is diluted at the population level by the large fraction of the cohort who never develop ESRD, I ran a second model conditioned entirely on ESRD onset. All individuals entered the model already diagnosed with ESRD in order to isolate the life-expectancy benefit of priority waitlist access for the sub-population who actually reach the waitlist. Entry to dialysis versus preemptive listing, dialysis and waitlist mortality, waitlist removal, post-transplant survival, and graft failure used the same age-stratified, all-cause parameters as the base-case model; no separate Healthy state or background-mortality term was included, since dialysis- and waitlist-state mortality rates are all-cause rates already measured in ESRD populations. I compared two arms, differing only in waitlist priority (priority vs. standard median wait, as above). The base case used an ESRD-onset age of 60 with an age sweep (40–75 years at ESRD onset) and one-way sensitivity analyses on wait times, per-cycle listing probability, preemptive-listing probability, post-transplant graft quality, and dialysis mortality ($\pm 20\%$).

Results

Base-case life expectancy

In the base-case analysis (age 40 at donation, overall population parameters, time-varying donor all-cause mortality HR as described in Methods), kidney donation was associated with a life-expectancy cost of $\Delta\text{LE} = -914.3$ days (2.50 years; donor 35.63 yr vs. non-donor 38.14 yr remaining life expectancy; Monte Carlo, 5×10^6 individuals per arm). The analytic cohort confirmed this estimate ($\Delta\text{LE} =$

–911.8 days). Across all base-case and subgroup scenarios, donation was associated with a net life-expectancy cost to the donor under this base-case HR assumption; no scenario identified donation as neutral or beneficial to the individual, with the exception of an alternative structural HR assumption examined below (“Structural sensitivity: donor mortality hazard ratio”).

Racial disparity

Race was a comparatively minor modifier of life-expectancy cost once the donor all-cause mortality HR, which affects all donors equally regardless of race, dominates the base-case result (Figure 2). At age 40, Black donors experienced a cost of 950.7 days, about 7% larger than the 892.5 days estimated for White donors. This gap was wider in relative terms at younger donation ages and closed entirely, with a slight reversal, at older ages: a Black donor at age 25 loses 1,404.0 days of life expectancy compared with 1,094.8 days for a White donor of the same age (a 28% difference), while at age 55 the two estimates are essentially identical and marginally reversed (586.7 vs. 588.8 days). This residual disparity traces to post-donation ESRD incidence: Black donors have a 15-year cumulative incidence of 74.7 per 10,000 versus 22.7 per 10,000 in White donors, ⁴ a $3.3\times$ difference further compounded by a higher background ESRD rate in age- and health-matched Black non-donors, ¹⁸ but this ESRD-driven gap is now a minor contributor relative to the much larger, race-neutral cost imposed by the time-varying mortality HR.

Age at donation

The life-expectancy cost decreased monotonically with older age at donation, from 1,131.6 days at age 25 to 591.2 days at age 55, roughly 91% larger at the youngest age relative to the oldest (Figure 2). Unlike the racial and sex disparities, this age gradient is driven primarily by the time-varying mortality HR rather than by ESRD risk: a younger donor lives long enough after the ten-year onset window for a much larger share of their remaining lifetime to fall under the elevated (up to 1.30) hazard ratio, whereas an older donor’s remaining life expectancy is increasingly concentrated in the years before the HR has fully risen.

Sex

Male and female donors faced broadly similar life-expectancy costs at age 40 (915.9 days for male donors vs. 888.3 days for female donors, a difference of about 3%), even though male donors carry a within-donor ESRD hazard ratio of 1.88 relative to female donors. ⁵ As with race, the sex-specific ESRD risk difference that dominated this comparison under a flat mortality HR is now a minor contributor next to the much larger, sex-neutral cost imposed by the time-varying mortality HR: a Black male donor at age 40 loses 971.3 days, about 6% more than the 915.9 days for an overall male donor and about 10% more than the 880.7 days for a White female donor of the same age (Figure 2). At age 25, Black male donors lose 1,506.8 days versus 1,048.4 days for White female donors, a 44% difference that is larger than at age 40 for the same

reason the racial disparity is larger at younger ages (see “Age at donation,” above).

Probabilistic sensitivity analysis

The PSA of 2,000 parameter draws (Figure 3), which holds the donor all-cause mortality HR fixed at its base-case time-varying profile (Methods), produced a mean Δ LE of -910.8 days (95% credible interval -936.1 to -889.3 days; $P[\text{donation beneficial}] = 0\%$). With the HR held fixed, the residual parameter uncertainty captured by the PSA (ESRD incidence, Weibull shape, waitlist listing probability) is narrow and consistently negative, in sharp contrast to the much larger uncertainty introduced by the choice of HR assumption itself (see “Structural sensitivity: donor mortality hazard ratio,” below).

One-way sensitivity analysis

With the donor all-cause mortality HR held fixed at its base-case time-varying profile rather than perturbed (Methods), no single parameter in the one-way sensitivity analysis produced a swing exceeding 42 days (Table 1, Figure 4). The largest remaining drivers were the donor ESRD relative risk (ranging from -867.8 days, $+41.5$ days vs. base, at $2\times$ controls to -932.3 days, -23.0 days vs. base, at the Mj  en upper-bound $11\times$) and dialysis mortality ($\pm 50\%$, ranging from -873.4 days, $+35.9$ days vs. base, to -927.3 days, -18.0 days vs. base). All other parameters, including PLD and standard waitlist wait times and post-transplant graft quality ($+19.5$ days at LDKT quality), shifted Δ LE by under 20 days. This is a direct consequence of removing the mortality HR from the sensitivity analysis: every scenario in Table 1 shares the same base-case HR profile, so none of them can reproduce the order-of-magnitude swings that the HR assumption itself produces (see below).

Structural sensitivity: donor mortality hazard ratio

Because the donor all-cause mortality HR is excluded from both the PSA and the one-way sensitivity analysis above, I separately compared the base-case time-varying HR profile against the flat HR implied by each individual published estimate, applied for the donor’s entire remaining lifetime (Figure 5). Taking the O’Keeffe et al. pooled point estimate (HR = 0.60) at face value and applying it for life, rather than treating it as evidence of no excess risk within the follow-up window it covers, reverses the sign of the result entirely: donation is associated with a life-expectancy benefit of 1,855.8 days. The textbook “no excess mortality” assumption (HR = 1.0 flat) yields a cost of only 68.3 days, an order of magnitude smaller than the time-varying base case. The base-case time-varying profile (1.0 ramping to 1.30 over years 10–15) yields -909.3 days. Applying the Mj  en et al. point estimate (HR = 1.30) from the moment of donation, rather than only after the ten-year lag their own data support, yields $-1,084.5$ days. The upper bound of the Mj  en et al. 95% CI (HR = 1.52), applied flat, yields $-1,694.0$ days. This 3,550-day range, from a multi-year benefit to a multi-year cost, dwarfs every other source of uncertainty examined in this analysis and is the dominant open question for any model of donor life expectancy.

Donor priority benefit

Although KAS250 affords prior living donors (PLD) priority access to the deceased-donor waitlist,¹⁴ the resultant life-expectancy benefit to the donor is small at the population level. Removing PLD priority access entirely, setting the donor waitlist to the standard 1,765-day mean wait, worsened Δ LE by 0.2 days relative to base case. Reducing the priority wait to 50 days (median, equivalent to a mean of ≈ 72 days) had negligible additional effect. This insensitivity to the exact priority wait time arises because fewer than 0.3% of donors develop ESRD within 15 years and fewer still ever reach the transplant waitlist; the priority pathway is simply too rarely invoked to move a population-level outcome meaningfully.

Voucher benefit

The life-expectancy benefit of a kidney-donation voucher to a healthy, non-donor family member peaked at $+1.2$ days for a recipient designated at age 25, falling to $+0.3$ days at age 40 and becoming negligible by age 60. These small figures reflect the same dilution mechanism: conditional on developing ESRD and reaching the priority waitlist, priority access adds a clinically substantial $+539$ days of life expectancy for a patient whose ESRD onset occurs at age 40, and $+109$ days for onset at age 60 (Figure 6). However, since only approximately 0.04% of non-donors develop ESRD within 15 years, this large conditional benefit compresses to well under one day at the population level. Voucher benefit was modestly higher for Black non-donor family members ($+1.5$ days at age 40), owing to their approximately four-fold higher baseline ESRD incidence.¹⁸

Discussion

The mean life-expectancy cost to the individual donor (≈ 911 days, or 2.5 years) must be weighed against the benefit conferred on the recipient and, through the voucher mechanism, on designated family members who may one day develop ESRD. The ESRD-conditional model demonstrates that priority waitlist access alone is worth more than 200 days to a patient who develops ESRD at age 55, and approximately 110 days for onset at age 60; the full benefit of receiving a functioning kidney rather than remaining on long-term hemodialysis is larger still. Viewed in this frame, donation still represents a transfer of life expectancy from donor to recipient, but on markedly less favorable terms than a smaller base-case estimate would suggest: under the base-case time-varying mortality hazard ratio, the donor forgoes a substantially larger share of their own expected remaining life than the population-level priority-access benefit alone returns, in exchange for the recipient’s prospect of years of additional survival.

This aggregate figure conceals heterogeneity, but less of it, and along a different axis, than a model driven mainly by ESRD-incidence differences would suggest. Age at donation is the demographic factor that matters most: the life-expectancy cost is roughly 91% larger at age 25 relative to age 55, because a younger donor lives long enough after the roughly decade-long delay in donor mortality risk for a much larger share of their remaining life to fall under the

elevated hazard ratio. Race and sex, which drove a three-fold and roughly 1.5-fold disparity respectively when post-donation ESRD incidence was the dominant cost driver, now differ by only about 7% and about 3% at the base-case donation age, because the ESRD-incidence differences that produce those disparities are a minor contributor once the much larger, demographically-neutral mortality-HR effect is accounted for. Age- and race-specific life-expectancy disclosure therefore both remain supported by this analysis, though on different grounds: a 25-year-old and a 55-year-old donor are, in effect, being asked to accept quantitatively different bargains under the same generic counseling framework, and a Black donor still faces a measurably larger cost than a White donor of the same age, most clearly so at younger donation ages. The racial gap is smaller than a model built around ESRD incidence alone would imply, and the possibility that donor mortality risk itself varies by race or by donor-recipient relatedness in ways this analysis could not evaluate (see Limitations) means the magnitude of the race-specific component should be treated as provisional rather than final, even where its direction is not.

The model also reframes what priority-access policy is actually protecting. KAS250 and the NKR voucher mechanism are designed around a rare event, an eventual ESRD diagnosis in a prior donor or designated family member, and this analysis shows that their unconditional, population-level life-expectancy value is correspondingly small (under one day for a voucher holder, under one day for donor PLD priority) simply because so few beneficiaries ever draw on it. This is not evidence that such policies are unimportant: conditional on the rare event they are designed for, priority access is worth 110 to over 500 days at the individual level (depending on age at ESRD onset), and the dilution reflects low incidence rather than low value per event. The appropriate way to communicate these mechanisms may therefore be closer to catastrophic insurance, whose expected value is small precisely because it is rarely invoked, than to an intervention expected to move average outcomes.

Several limitations temper these conclusions:

- The results are dominated by a single structural uncertainty, the time-course of the donor all-cause mortality hazard ratio, rather than by sampling error in any individually estimated parameter. Short-to-medium-follow-up cohorts and the meta-analysis pooling them report no excess mortality, while the one cohort followed long enough to plausibly detect a late-emerging effect reports a hazard ratio of 1.30 that its own authors attribute to risk emerging only after roughly a decade. This analysis encodes that pattern as a piecewise-linear ramp (flat at 1.0 through year 10, rising to 1.30 by year 15, matching the Mjøen et al. cohort's median follow-up), which is a minimal, directly literature-motivated functional form but an assumption nonetheless; the true time-course, and whether it is linear, could differ. Unlike most parameters in this model, this uncertainty is not confined to magnitude: taking the alternative literature point estimates at face value for the donor's entire life ranges from a multi-year life-expectancy *benefit*

($HR = 0.60$) to a cost exceeding four years (the upper 95% CI of the long-follow-up estimate), so neither the size nor the sign of the donor's life-expectancy cost should be considered settled. The true value likely also depends on donor-recipient relatedness and selection practices that differ across health systems and were not modeled here.

- The model reports only undiscounted life-years and does not incorporate quality-of-life weighting; because dialysis is associated with substantially reduced health utility, a quality-adjusted life year (QALY)-adjusted analysis would likely widen, not narrow, the donor's disadvantage during any dialysis-dependent interval.
- Several parameters were necessarily borrowed across source populations under an explicit sourcing rule (e.g., within-donor hazard ratios from Massie et al. applied to the Muzaale matched-control baseline), and no single cohort traces the complete pathway from living donation through ESRD onset to transplantation, so the calibration checks reported here constitute face validity rather than external validation against a purpose-built dataset.

Future work should prioritize prospective donor cohorts with follow-up long enough, ideally beyond 15–20 years, to directly estimate whether and when donor all-cause mortality risk begins to rise, since this single question now determines whether donation carries a life-expectancy cost at all.

Acknowledgments and Statements and Declarations

Ethical Considerations

This study analyzed only publicly available, de-identified, aggregate data from published cohort studies and national registry reports (USRDS, SRTR/OPTN Annual Data Reports); no primary human-subjects data were collected. Institutional Review Board approval was therefore not applicable.

Consent to Participate

Not applicable. This study did not involve primary data collection from human participants.

Consent for Publication

Not applicable.

Declaration of Conflicting Interests

The author is a living kidney donor; that experience motivated this study. The author declares no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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Data Availability

All parameter sources are published, publicly available studies and registry reports, cited throughout the manuscript. The simulation code, parameter-assembly pipeline, and calibration checks are implemented in Python 3.10 (NumPy, pandas, matplotlib) and openly available in a public repository at <https://github.com/rupumped/kidney-donor-le>, consistent with Level 1 of the Transparency and Openness Promotion (TOP) Guidelines. No restricted-access data were used.

Demographic Classification Variables

Race is used in this analysis as reported in the underlying data sources rather than as a directly measured biological quantity. Donor-side race data^{1,4,5,19,20} derive from categories reported to the Organ Procurement and Transplantation Network (OPTN) at donor evaluation and registration; the non-donor comparator population used by⁴ was drawn from the Third National Health and Nutrition Examination Survey (NHANES III), in which race is self-reported by participants. Race is included in the model because it is the variable available in these source populations and because it is strongly associated with measured post-donation ESRD incidence. Part of this association has an identified biological correlate, *APOL1* renal-risk-variant genotypes, which are more common among individuals of recent African ancestry and are directly implicated in donor kidney-disease risk.¹² However, socioeconomic status, healthcare access, comorbidity burden, and other factors correlated with self-reported race in the United States were not available in the source registries and are not separately modeled here. The race-stratified life-expectancy estimates reported in this paper should therefore be interpreted as reflecting the net effect of these unmeasured, race-correlated factors together with any effect of race itself, rather than race operating as an isolated causal variable.

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Tables

Figures

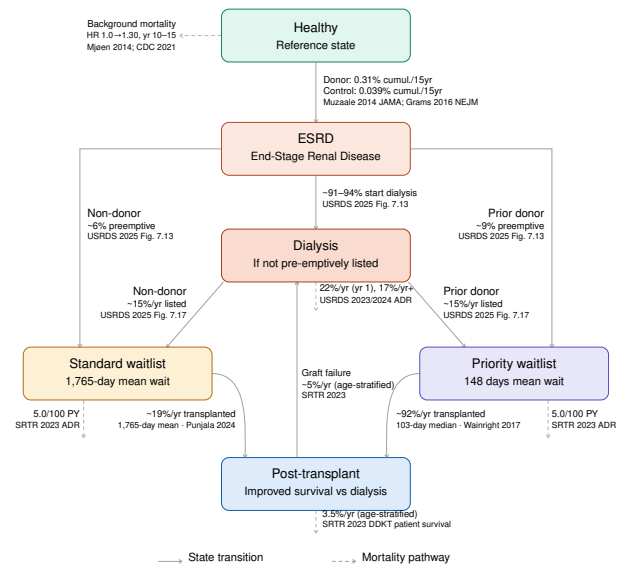


Figure 1. Markov state-transition model for kidney donation. Boxes represent the seven mutually exclusive health states; arrows show permitted transitions. The donor arm (upper pathway) routes ESRD-onset patients to the Priority waitlist; the non-donor arm routes to the Standard waitlist. The Dead state is absorbing.

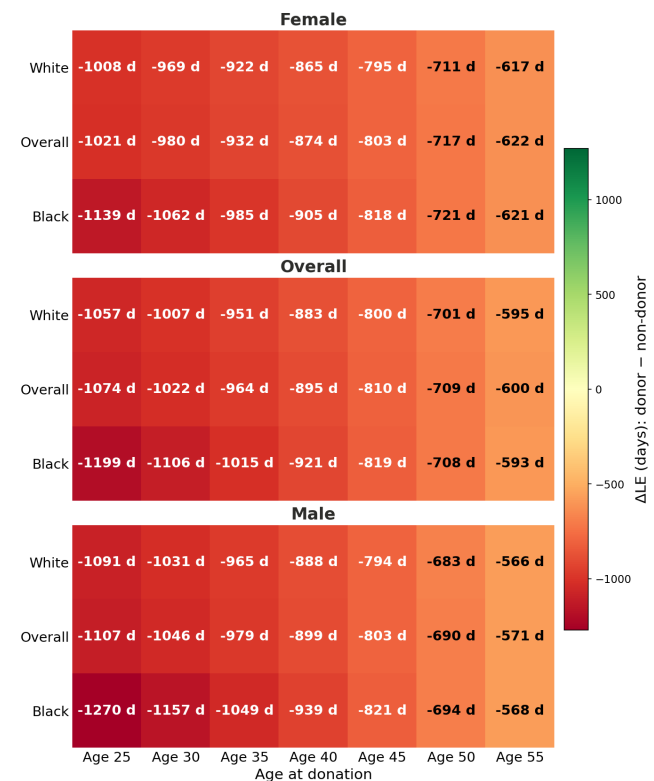


Figure 2. Life-expectancy impact of kidney donation by age, race, and sex. Each cell shows ΔLE (donor minus non-donor, days) for the indicated sex panel, race row, and age-at-donation column. Age at donation dominates: a Black male donor at age 25 loses 1,507 days, compared with 619 days for a White female donor at age 55. Racial and sex differences within an age group are comparatively small: at age 25, the largest within-age contrast (Black male vs. White female) is 1,507 vs. 1,048 days, a 44% difference.

Table 1. One-way sensitivity analysis: parameter values tested and the resulting change in life expectancy (donor arm minus non-donor arm), age 40 at donation, all other parameters held at base-case values, including the donor all-cause mortality HR, which is held at its base-case time-varying profile rather than perturbed here (see Table 2 and Figure 5 for sensitivity to that assumption). Negative Δ LE values indicate a net life-expectancy cost of donation relative to the matched non-donor arm under the value tested; positive values indicate a donor-arm benefit. The “8× controls (base case)” row reproduces the overall base-case result reported in the main results (Δ LE = −909.3 d).

Parameter (value tested)	Δ LE (days)	vs. base (days)
<i>ESRD relative risk, donor vs. non-donor (base case: 8×, 30.8/10,000)</i>		
2× controls	−867.8	+41.5
4× controls	−880.4	+28.9
8× controls (base case)	−909.3	0.0
11× controls (Mjølén upper bound)	−932.3	−23.0
<i>Standard waitlist mean wait (base case: 1,765 d)</i>		
24 months (730 d)	−909.3	0.0
61 months (1,857 d, pre-KAS250)	−909.3	0.0
<i>Priority (PLD) waitlist mean wait (base case: $\approx$ 148 d, from 102.6 d median via $\mu = \text{median} / \ln 2$)</i>		
50 d median ($\approx$ 72 d mean, optimistic)	−909.3	0.0
200 d median ($\approx$ 289 d mean)	−909.4	−0.1
No priority access (PLD = standard, 1,765 d mean)	−909.5	−0.2
<i>Dialysis annual mortality (base case: 17%)</i>		
−50% (8.5%)	−873.4	+35.9
+50% (25.5%)	−927.3	−18.0
<i>Post-transplant survival (base case: DDKT quality)</i>		
LDKT quality (SRTR Figure KI 76)	−889.8	+19.5

Table 2. Structural sensitivity: donor mortality hazard ratio. Δ LE under the base-case time-varying HR profile versus a flat HR taken from each individual published estimate and applied for the donor’s entire remaining lifetime, age 40 at donation. Unlike Table 1, this range reflects a genuine disagreement about model structure (whether, and when, donor mortality risk rises) rather than sampling uncertainty.

HR assumption	Δ LE (days)
HR = 0.60 flat (O’Keeffe et al. 2018 pooled estimate)	+1,855.8
HR = 1.00 flat (no excess mortality)	−68.3
HR 1.00 $\rightarrow$ 1.30, years 10–15 (base case)	−909.3
HR = 1.30 flat (Mjølén et al. 2014 point estimate)	−1,084.5
HR = 1.52 flat (Mjølén et al. 2014 upper 95% CI)	−1,694.0

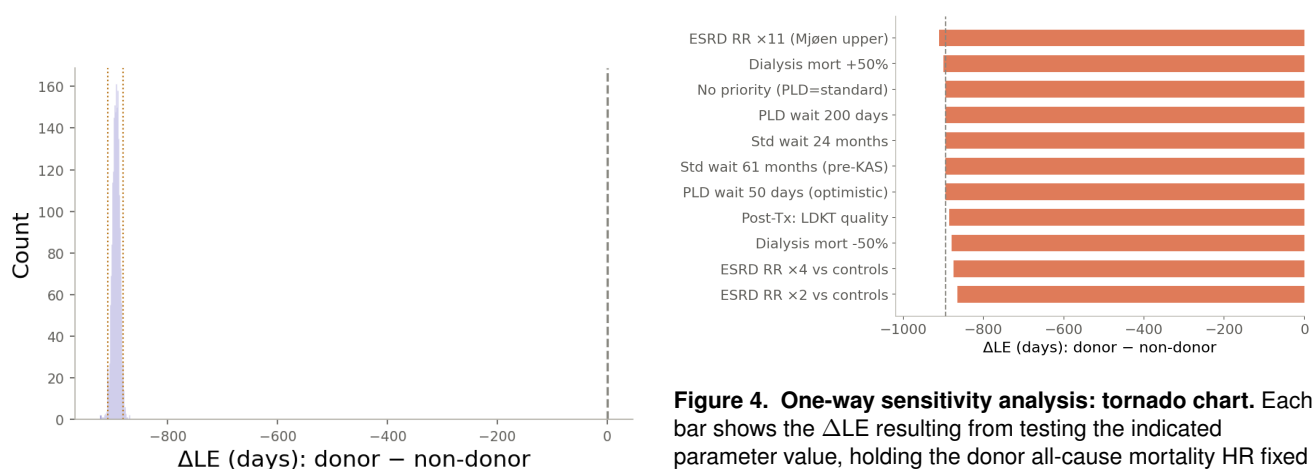


Figure 3. Probabilistic sensitivity analysis (PSA). Histogram of Δ LE (donor minus non-donor, days). The dashed vertical line marks zero; orange dotted lines mark the 95% credible interval bounds.

Figure 4. One-way sensitivity analysis: tornado chart. Each bar shows the Δ LE resulting from testing the indicated parameter value, holding the donor all-cause mortality HR fixed at its base-case time-varying profile (see Table 1 for exact values; Figure 5 for sensitivity to the HR assumption itself). Vertical dashed line shows base case. With the HR held fixed, no parameter shifts Δ LE by more than 42 days; priority waitlist parameters (“No priority” and “PLD wait” rows) are barely distinguishable from base case, confirming the negligible population-level effect of transplant priority.

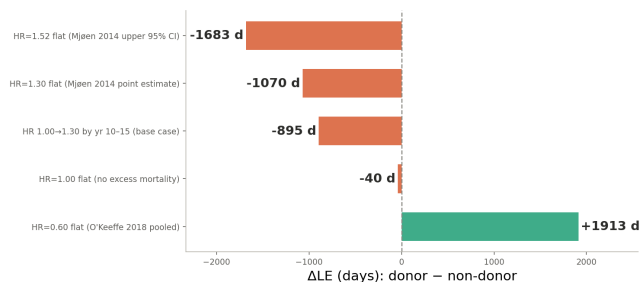


Figure 5. Structural sensitivity to the donor all-cause mortality HR. Each bar shows the Δ LE resulting from replacing the base-case time-varying HR profile (1.0 flat through year 10, ramping to 1.30 by year 15) with a single flat HR held for the donor's entire remaining lifetime, taken from an individual published estimate. Vertical dashed line marks zero. The choice of HR assumption spans a 3,550-day range, from a multi-year life-expectancy benefit (HR = 0.60, O'Keeffe et al. pooled estimate) to a cost exceeding four years (HR = 1.52, upper bound of the Mjoeen et al. 95% CI), dwarfing every other source of uncertainty examined in this analysis.

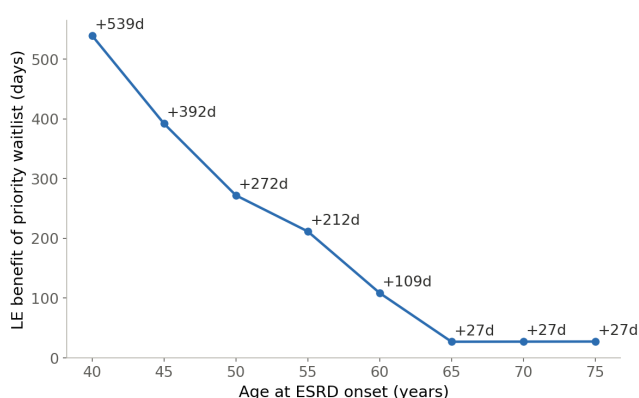


Figure 6. Priority waitlist LE benefit conditional on ESRD onset, by age. Δ LE (priority minus standard waitlist, days) for individuals who have already developed ESRD, as a function of age at ESRD onset. Conditional on ESRD, priority access is highly valuable: +539 days at age 40 onset, +109 days at age 60. This large conditional benefit is diluted to under 1 day at the population level by the low ESRD incidence in non-donors ($\approx 0.04\%/15$ yr) and in donors ($\approx 0.3\%/15$ yr).